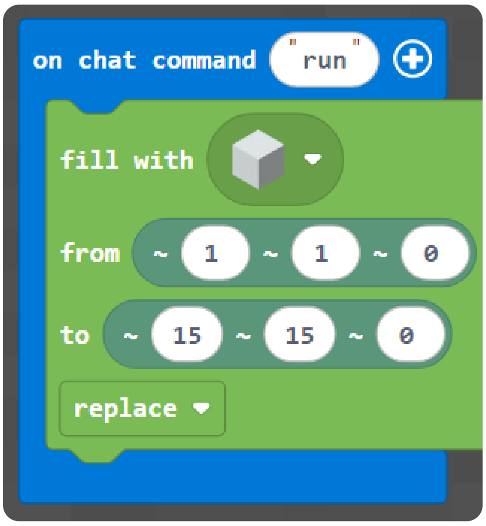
M003 Extension — English Worksheet (Intermediate)

Topic: Pixel-Art Mobs (x, y) · **Course:** 3D Coordinates · **Level:** Intermediate (Extension, after Week 3) · **Time:** about 40 minutes

This is a **bonus picture challenge** between Week 3 and Week 4. You already copied one small picture with **(x, y)**. Now you copy a whole mob — a **pink axolotl** — onto your wall. Every block sits at a spot named by **two numbers (x, y)**. x is how far **across**, and y is how far **up**.

First, build your canvas. Run this to make a blank **15 × 15** wall, then put a **red block at your feet** as your **home spot**:

Blocks



Python

```
def on_run():
    blocks.fill(WHITE_CONCRETE,
        pos(1, 1, 0),
        pos(15, 15, 0),
        FillOperation.REPLACE)
player.on_chat("run", on_run)
```

> 🔴 **Big idea:** (x, y) is counted from your **home spot** — not from the corner of the world. Stand on the red block *every* time you run, or the whole picture shifts. Move your feet, move your picture.

1 · Predict

Read each set of steps. Write a full sentence about what you think you will see on the wall.

```
place pink block at (4, 13)
place pink block at (5, 13)
place pink block at (6, 13)
```

Three pink blocks share y = 13. What shape do they make, and which number stays the same?

```
place black block at (8, 7)
place black block at (8, 8)
place black block at (8, 9)
```

These three share x = 8. What shape do they make, and which number changes?

```
place pink block at (6, 10)
place black block at (6, 10)
```

Two blocks aim for the same (x, y). What do you see — pink, black, or both? Why?

2 · Spot the Bug 🐛

Each block of code is broken. Read what it should do, write the fixed code, then explain in one or two sentences why the original was wrong and why your fix works.

Bug A — This should make the axolotl's two eyes, one at (6, 10) and one at (11, 10). Right now both land in the **same spot**.

```
place black block at (6, 10)
place black block at (6, 10)
```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — The dragon's eye should sit at (8, 10). Right now the two numbers are **swapped**, so it lands in the wrong place.

```
place purple block at (10, 8)
```

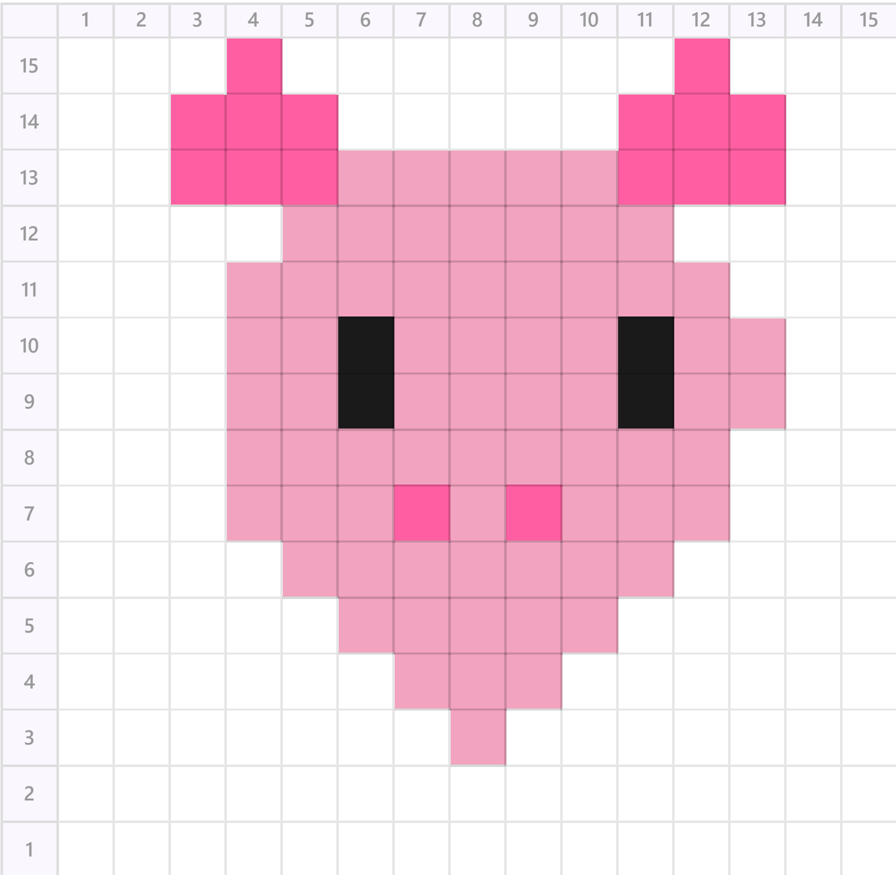
Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Build It 📸

Stand on your **home spot** (the red block). Build the 🦜 axolotl below onto your wall, block by block, by reading the (x, y) of every colored square. The numbers across the **top** are **x**; the numbers down the **side** are **y**.

🦜 Axolotl (pink · #f2a3c0, frills · #ff5fa2, eyes · black)



When your picture is finished, send a photo of it, then explain what you did. Use these sentence starters — write about 4 sentences total.

First, I built the …

The lowest block in my picture is at (x = …, y = …).

To go across a row I changed the … number; to go up I changed the … number.

One time my picture landed in the wrong place because …

4 · Explain It 🎥

Take a video on your phone (or a parent's phone) while you show your axolotl on the wall. Talk through it like you are teaching someone who has never seen it. Try to use these words: **x**, **y**, **coordinate**, **across**, **up**, **home spot**.

> 1. Show your home spot (the red block) and say why you stand on it every time.
> 2. Show your finished axolotl from the front.
> 3. Point at one block and read its coordinate out loud: "this one is at x ..., y ...".
> 4. Show one block that you placed in the wrong spot at first, and how you fixed it.

Write about 4 sentences for your video. Plan it here before you record — you can read from it while filming.

Submit ✅

Send this worksheet + your walkthrough video to teacher on KakaoTalk.